

Harshit Periwai

NYU Admission Profile & Application Summary

Course Applied For - Computer Science / Artificial Intelligence

Acceptances

1. Penn State University + Schreyers Honours college + \$10,000
2. University of Massachusetts Amherst + \$18,000
3. Illinois Tech + \$20,000
4. Rochester Tech + \$30K + Accelerated Masters
5. Queen Mary University London
6. Newcastle University + €7000
7. Michigan State University + \$17,000
8. University of California Santa Barbara
9. New York University + Global Gateway
10. Northeastern University
11. University of California Davis

Waitlists

1. University of California Irvine
2. University of California San Diego

Rejects

1. UIUC
2. Purdue University
3. Columbia University

Common Application

Honors in Order

1. **Principal's Scholar - Grade 11 and Grade 12**
Grades 11 & 12, School Level
2. **CS Merit and Class Topper**
Grade 11, School Level
3. **Gold Crest Award Recipient**
Grade 12, International Level
4. **Sanskrit Merit**
Grades 9 and 10, School Level
5. **Future Tech Olympiad - All India Rank: Top 50**
Grade 11, National Level

Grades and Testing

Board - CBSE

Grade	Percentage (100 Scale)	IELTS Results Reading: 8.5 Writing: 6.5 Speaking: 8.5 Listening: 9.0 Overall Band Score: 8.0
9th	84%	
10th	95%	
11th	74%	
12th	80% - 85%	

SAT

Section	Score
Evidence Based Reading and Writing	680
Mathematics	750

Extracurriculars in Order

1. Research Paper on AI literacy in rural India

Surveyed villagers about AI challenges using anonymous, multilingual forms. Led the AI Literacy Drive to bridge the technology gap.

2. Eco Bin - AI Powered Waste Segregation App

Built an offline AI app to classify wet, dry, and e-waste using the phone camera. Trained on Kaggle data and deployed in Unity with community outreach.

3. AI Literacy Drive

Created a multilingual AI course certified by Nice Software Solutions (Nagpur) and delivered online and in nearby villages.

4. Engineering Intern, Lighthouse Learning

Led backend interns to build the Future-You AI feature, moving from ComfyUI to scalable standalone code. Now live on Argus App.

5. Model United Nations

Won top prizes in 5+ MUNs. Led CENMUN promotions, built cenmun.com, and served as Director General for GLOMUN, raising Rs.20K+ for charity.

6. AI Captain: School Technology Integration Leader

As AI Captain, led a tech support team to fix 50+ issues and help 30+ teachers, creating resources to support adaptive learning.

7. Youtubing

Built a 4K+ subscriber YouTube channel, pioneered modding, and collaborated with youtuber Young Fro (100K+ subscribers).

8. Data Science Intern, Nice Software Solutions (Nagpur)

Contributed to AI projects by preparing datasets, training and optimizing models, and supporting deployment in client-facing applications.

9. Tennis, Lawn Tennis Player

Played lawn tennis since Grade 5 and competed at district level for three consecutive years. Namely, in grade 9, 10 and 11.

10. Created Garry's Escape

Created "Garry's Escape" a 2D physics-based educational game that became state champion and national runner-up in the National Design Championship '23.

The Essay

Describe a topic, idea, or concept you find so engaging that it makes you lose all track of time. Why does it captivate you? What or who do you turn to when you want to learn more?

-> I entered the generated CVV and the transaction went through successfully. My eyes twinkled with joy. This was the 11-year-old me trying to start an Amazon Prime trial without owning a legitimate credit card. I did not know whether my actions were legal or ethical; for me, everything came down to whether the method actually worked. Fortunately, I quickly realized my mistake. It became a one-time childhood experiment I never repeated.

Technology continued to shape my childhood, from exploring the golden era of mobile games to building projects that eventually won national awards. That same spirit of curiosity and challenge carried into my gaming journey. After playing one mobile game for nearly a year, I became eager to explore its possibilities beyond the original design. While searching for a modded version, I came across YouTubers claiming the game was impossible to modify. That claim ignited my determination. I treated their words not as a fact but as a challenge.

I took on the challenge and developed a working mod using Unity and C#, which I posted on YouTube. To my surprise, the video garnered over 10,000 views. This was the start of my YouTube journey. As I built a community of over 4,000 followers, I not only polished my game development skills but also learned about content creation.

Along with these learnings came an understanding of the responsibility that accompanies technology. While my early experiments were driven by curiosity and a desire to see what worked, over time I began to realize there was a huge gap between technology and its users. One day, I noticed an issue: when the community helpers arrived to collect waste in two separate bins, wet and dry, residents had trouble identifying which waste went where. I conducted a survey that received over 100 responses from people from all walks of life, and the results were surprising. It was clear that awareness was needed. My journey of exploration and experimentation took a turn toward real-world issues.

To solve this problem, I developed an AI-enabled app to help users sort waste effortlessly. I conducted 30+ awareness sessions with community helpers, training them to use the Ecobin app for segregating dry, wet, and e-waste. Within two months, over 100 households began using the app regularly, leading to visible improvements in waste segregation, as vouched for by community helpers.

Seeing the acceptance and adaptation of technology, I realized that if users are taught how to use it properly, major changes in their lives can take place. All this led to my most passionate movement to date, the AI Literacy Drive. Many people were unaware that the power of AI already existed in their smartphones. This movement became my way of sharing knowledge, so through a series of short, self-produced YouTube videos, I simplified concepts like neural networks and ethical AI using real-world analogies. The videos drew over 2,000 views within a short span of time, and 70% of viewers found the explanations easier to grasp than typical online resources. Community members, including local students and working professionals, reached out with questions and examples of how they applied what they learned, from identifying AI in their daily apps to building basic models. This makes one thing clear: people don't need money or complicated machines. The only thing required is curiosity, the same curiosity that brought me this far. For me, this movement was about empowerment, curiosity, and the sheer joy of seeing individuals realize that they already possess great power in their pockets.

Today, and in the future, my motto will always remain the same: technology is best when it is harnessed for the betterment of mankind and nature. The thrill and joy on people's faces when they interact with new technology take me back to my childhood days of exploration. I want to bring the same emotional and real-life impact to as many people as I can.

Additional Information

Personal Website: <https://harshitperiwal.in> Please visit my website to try out my AI projects and games and view my resume. You may also try HarshitGPT, an LLM fine tuned by me, and ask it anything about my activities. Completed a course on Number Theory and Cryptography by Michael Levin (UCSD). Fully sponsored attendee of the IGDC (Indian Game Developers Conference) where I met and took insights from industry leaders.

Penn State University (56%)

Academics and Interests

- . **Type of Degree:** Baccalaureate (4-year)
- . **Intended Major/Program of Study:** Artificial Intelligence Engineering, B.S.
- . **End Campus:** University Park - main campus
- . **Start Campus:** University Park - main campus

Schreyers Honors College Essays (8%)

Essay 1

Artificial intelligence, as of now, is largely monopolized by a handful of big technology powerhouses that not only have the resources but also the control over the data, computing power, and infrastructure. Essentially, these companies, by virtue of their joint ventures, create and sustain closed loops that demand gargantuan money for any radical innovation to emerge from outside these loops. In order for startups to have access to the computation, they must, quite often, already have hefty capital. A person intending to operate AI locally is required to buy costly gadgets and equipment, whereas a user of the cloud is trapped in a subscription model with payment installments that are bound to increase gradually. The cause of this malfunction is not software, but hardware. To maintain the world economy, a kind of dependence on semiconductor manufacturers has been developed, which at the same time, is quite hazardous as well, as these few major companies significantly influence the whole market of high tech. For instance, TSMC is the company that mainly produces chips that are utilized by NVIDIA, Apple, and other tech companies. Such concentration of power in few hands creates scarcity and very high prices for powerful hardware and also makes the system more vulnerable since it links the source of essential facilities to a small number of geographical regions, thus, if the supply chains are broken, the whole system will be at a standstill. When smart systems are tied to such supply chains, then technology is no longer a common right but becomes a privilege. It impacted the way I understand the world. When I was exploring AI and machine learning, I was constantly restrained by my hardware capabilities since my laptop did not have a GPU. I was only able to perform minimal training of models and had little freedom in experimentation. Similar situations came up in my work with Blender, where the limitation was not my creativity but the hardware that I had. During my internship at Lighthouse Learning, I was confronted with the same problem. The developers on the team I was working with were facing hardware limitations, thus they could not perform AI models running locally. Consequently, they had limited access to testing alternatives, performance comparisons, and most importantly, free experimentation during development. Therefore, using this inefficient method, innovation slowed down and the effectiveness of technical decisions lessened. My great transformational thought is about minimizing the world's reliance on a centralized chip ecosystem controlled by a monopoly through the redesign of AI systems which are capable of operating on decentralized hardware that can be widely manufactured. Not doing away with GPUs altogether, but rather refusing to accept the notion that the development of powerful AIs has to be based on a few

rare, cutting edge silicon chips, is what this means. Considerations such as the accessibility, resilience, and efficiency of the AI system instead of mere raw power ought to be the focal point. From a technical perspective, the goal is to develop hardware independent AI models that are executable on a CPU, mobile chips, edge devices, or even older semiconductor nodes. The focus would be diverted from the infinite scaling of model size to areas like compressed architectures, task specific systems, sparse computation, and local first inference that is made possible by methods such as federated learning and quantization. I ran an AI Literacy Drive through which I came to understand very well that even though there are many talented people out there, they are still deprived of the use of powerful tools due to high costs and lack of access. By lowering the cost of AI and spreading the hardware, it would not only be a straightforward removal of artificial barriers but also a widening of the potential of such a change, if brought about, would make AI available to the general public like any other utility rather than being exclusive to a certain class of corporate elites. The power of data would then be handed over to individuals, tools would no longer be the monopoly of privileged schools and startups, as they will have real access to them, and countries will be able to lessen their dependence on fragile supply chains. The geographical points from where innovation will be possible will thus be more rather than fewer. On the grand scale, I have the intention of creating such systems that see intelligence as a universal capability, just like breathing, and not as a limited exclusive country resource. Limited hardware experiences, professional development projects, and leadership in AI education are the factors that have molded this idea of mine.

Short Answer 1

My intellectual and personal growth has been shaped by curiosity, cultural values, and community driven experiences. My inquisitive mind led me to explore technology from a very young age through self-taught C#, Unity, 3D modelling in Blender, and game modding. Growing up in a Rajasthani family, I learned that innovation comes from resourcefulness, that is, doing more with less. This later became the foundation of how I think and solve problems, and has strengthened my confidence and creative logic. Through EcoBin and AI Literacy Drive, I have seen that real change begins when people feel involved and supported, not just instructed. As the AI Captain of my school, I helped more than 30 teachers, for the integration of iPads for juniors, through the techbuddy program. All these events made me believe that technology should empower people and leadership is about teamwork and service. I got recognition from the YouTube community for game modding and my collaborations with larger creators validated my efforts and encouraged me to keep experimenting with ideas. Together, these values and engagements have made me a curious creative thinker, resourceful and driven to build technology that creates real impact for everyone.

Short Answer 2

Did you know nearly 67% of Japan is covered with forests, and one of the major contributors is the Miyawaki method, a technique used to create thick forests from native species? Japan masters the art of balance. Japanese culture has immense respect for nature while simultaneously building futuristic cities. This is the reason why I am drawn to Japan. I would love to visit Tokyo, where neon lights on flashy billboards, skyscrapers, and high-speed transport make it feel like a city of cutting-edge technology. It is like visiting the future, where creativity and technology shape everyday life. On the contrary, Oshino Hakkai, a village near Mount Fuji, is completely the opposite. In this village, silence is serene, life is simple, and nature is present in every direction. What fascinates me is how Japan blends both worlds together effortlessly. It doesn't choose between tradition and growth; it preserves forests while building ultramodern cities. It celebrates modernity without forgetting its roots. The harmony between vibrancy and calm, bonsais and skyscrapers, neon and nature is why Japan is my favorite destination. This is the kind of future I want to build, where growth and traditions go hand in hand.

Short Answer 3

Over the last four years, my extracurricular journey has revolved around leadership, technology, and community impact. These activities have mostly involved serving as the head website designer for MUN promotions, interning at Nice Software Solutions, conducting a research study on AI literacy in rural India, and creating an AI Literacy Drive through educational videos. Each activity has strengthened a different part of me. One such experience that has shaped me is my Lighthouse Learning Internship, where I led a team of 10 interns from schools across India. I was selected for the internship because of my role as the AI Captain of my school, a position where I helped students and teachers with new technologies. At Lighthouse Learning's head office in Pune, my team and I worked on developing a new feature for the Argus school app called "Future Me." The feature allows students to visualize themselves in the profession they aspire to be, using an open-source stable diffusion model and utilizing the CUDA framework. Leading a team with diverse skills taught me responsibility, decision-making, and the importance of communication. This internship tested all my skills in one go, including leadership, initiative, technical understanding, and teamwork in a real corporate environment.

Short Answer 4

When I built my waste-segregation app, I thought technology would change communities, but I was wrong. It is the people who change communities, and technology only helps them start. I made EcoBin with a simple thought: to help families separate dry and wet waste using a multilingual, offline app. However, when I introduced it, people were hesitant. The elderly were not comfortable using apps, parents did not want an extra step, and some did not understand the importance of segregation. Nothing changed. Instead of forcing the current system, I tried something different. I turned to the most openminded group in society: children and teenagers. I asked them to take a selfie with their properly sorted waste and post it in the society WhatsApp group. This started a chain reaction. The group was full of kids holding green and blue bins. Shortly after, grandparents joined with enthusiasm, followed by parents. Within a span of two months, more than 100 families started using the app freely. EcoBin took the shape I intended it to take: it became a community movement. It taught me that leadership is not about convincing, it is about inspiring.

Short Answer 5

Over the last four years, the recognitions I have received have motivated me to perform better. One of the most meaningful was the Principal's Scholarship, which is awarded for strong academic excellence in tenth grade. I scored 95%, comprising A's in all subjects. I also earned a Top 50 All-India Rank in the Future Tech Olympiad, an examination organized by 1M1B and IBM, which tests students for their knowledge of the latest technology and machine learning. Scoring among the top students nationally strengthened my confidence. I was the National Runner-up for design championship in the game design category for my game Gary's escape, this win got me to attend the biggest game developers conference in Hyderabad. Another recognition was the British Science Association CREST Awards. I won the Silver Award for my app EcoBin, which is a waste segregation offline app that uses a model trained on Google Teachable Machine, but was encouraged by evaluators to reapply for the Gold level. I declined the Silver and resubmitted my upgraded work for Gold; waiting for results. This experience taught me to be persistent and work toward a higher standard. Together, these recognitions have shaped my confidence and interest in science and technology.

Short Answer 6

The recognition that changed my outlook was winning the National Runner-Up at the Design Championship competition for the game design category. Winning the award was very

prestigious, but what came after mattered the most to me. I was invited to the gaming industry's biggest conference, IGDC (Indian Game Developers Conference), in Hyderabad to showcase my game. This was my first exposure to the professional game development world. I met developers, studios, and technology creators all under one roof. I understood that game design was far bigger than any project I had built at home. There were many seminars organized at the conference. I attended a few of them to understand the latest tools and trends in the industry and explored how real studios function. I also met creators like Nikhil Malankar, whose expertise in game development opened my eyes to how the industry is booming. This experience had a strong impact on my outlook. It made me understand that learning never ends after a project is completed, but in fact, it begins when it enters the real world and you see how it works. It sparked curiosity, and I learned that there is more to grow, build, and innovate.

Short Answer 7

I was barely ten when I first heard the line, "With great power comes great responsibility," in the Spider-Man Raimi trilogy movie. At the time, it meant making sure I did not injure myself while trying to be Spider-Man, swinging from the chair to the sofa. I was innocent, adventurous, and completely opposite of what Uncle Ben actually meant. As I grew older, I began to perceive Spider-Man differently. His journey is about learning to balance ambition, relationships, and values. He juggles school, friendships, and a city full of problems. This made me realize that responsibility is not easy, especially when it is not rewarding. What I admire most is his humility. He saves the city and still worries about Aunt May like a normal human being. This helped me understand that being grounded is more important than being glorified. I am not a superhero who can save the world, nor did Spider-Man give me any special powers. But he taught me that kindness, humility, and responsibility are the real superpowers on which I can build my future, and if not save the world, I can at least bring about a positive change in it.

Short Answer 8

I belong to Rajasthan, one of India's driest desert states, where food and water are scarce. From this dearth, Rajasthanis build abundance. For us, "less is more" is a survival instinct passed down through generations. Growing up, I watched my grandmother manage things so efficiently that nothing ever went to waste. The lesson I learned was that real innovation lies not in having more but in doing more with less. Over the past four years, I have seen that same guiding principle reflected in AI. A few words can complete hours of work in minutes. For me, AI is a digital extension of my Rajasthani cultural lineage, and it motivates me to use technology to empower people. This perspective has changed my approach to all the projects I have worked on, like EcoBin for sustainability, which helped over 100 households segregate waste, Apple iPad integration at school for tech support and cost cutting for over 30 teachers, and an AI literacy drive for empowerment and growth through videos for interested learners. I want to channel my unique heritage with technology to design systems that waste less, serve more, and work smartly.

University of Massachusetts Amherst (60%)

Major: Computer Science

Why this Major?

I chose Computer Science because I've always been fascinated by logic, patterns, and how things are built. Maths and Newtonian physics, built on vectors and calculus, showed me how complex systems follow simple rules. This drive fueled my Top 50 All-India Rank in the Future Tech Olympiad by 1M1B and IBM, which tests technology and machine learning. During my internship at Lighthouse, I developed a feature called "Future Me" for our schools' app - Argus. It uses an open source stable diffusion model to help students visualize their aspirational career. These experiences made CS my language to explore, question, and design.

Why UMass Amherst?

UMass's call to "Be Revolutionary" reflects a responsibility I want to embrace to overcome barriers rather than accept them. As I learned AI, I realized our progress depends on a fragile semiconductor ecosystem, where few manufacturers control price and accessibility. I experienced this limitation in my AI projects and internship at Lighthouse, where limited hardware held developers back due to elusive tools. The Early Research Scholar Program at CICS reinforces my belief that UMass is the ideal place to begin. Under the guidance of Professor Subhanshu Maji, I am eager to contribute to the Computer Vision and Research Lab.

Narrative Community

I belong to Rajasthan, India's desert state. Growing up in a Rajasthani family, I learned that innovation comes from resourcefulness, doing more with less. I saw this principle reflected in technology and artificial intelligence. A few words could complete hours of work. I used AI to help communities segregate waste with Ecobin and made technology accessible for teachers and students with my Apple iPad Integration initiative. I want to channel my heritage with technology to design systems that waste less, serve more, and work smartly. I bring this traditional, successful approach to building a future where less truly becomes more.

University of California – Personal Insight Questions (PIQs)

PIQ 1: Describe an example of your leadership experience in which you have positively influenced others, helped resolve disputes, or contributed to group efforts over time.

When our school introduced iPads for fourth and fifth graders, all students and teachers were excited. However, it brought a fair share of challenges. Teachers and students alike struggled to open websites, scan QR codes, and perform basic tasks. The difficulty lay in the differences between Apple and Android platforms, which they were more comfortable with.

As the inaugural AI Captain of my school, I felt responsible for building a positive culture around this. I was selected as the AI Captain in Grade 11 by the principal of my school. Even before getting the role, I was known for my ability to use AI among students in my class and juniors. I started working on iPad settings and exploring app integrations. I created simple, step by step image depictions to fix common issues, like how students often could not connect to the Wi-Fi needed for online quizzes.

I made sure all my projects were open source and was happy to explain my logic and code. In this way, I was able to create groups of peers who, in turn, could help more students. I formed an iPad Buddy Team, a small group of tech savvy students who could assist in every class. Together, we created resources that included troubleshooting videos, how to sheets, and answers to frequently asked questions.

In one month, the iPad Buddy Team solved over fifty recurring technical issues and helped more than thirty teachers. It was nothing short of revolutionary to see the initial frustration and drop in morale give way to renewed energy and commitment to adaptive educational technologies. Our teachers began experimenting with interactive lessons and digital quizzes. Entire classrooms began scanning QR codes and opening synchronized sites effortlessly.

I began this journey of leading with a sense of personal responsibility but ended up being awed by the power of collective effort. Sundar Pichai (CEO, Alphabet Inc) once said that the future of AI is not about replacing humans; it is about augmenting human capabilities. I think that has been my experience of leadership as well.

PIQ 2: Every person has a creative side, and it can be expressed in many ways: problem solving, original and innovative thinking, and artistically, to name a few. Describe how you express your creative side.

I was taking my Dermatoglyphics Multiple Intelligence Test, a test to understand my strengths and learning style. The results were different from what I had imagined. It identified me as a "creative thinker." I could not believe it. "I do not draw or paint," I said. The counselor replied,

"Creativity is not limited to art; it can be logical too." I wondered and thought, how is that even possible?

That conversation perplexed me and left a lasting impact. A few weeks later, I was playing The Wrestling Empire game on my laptop. It is a wrestling simulation game. I discovered that it could be modified. I began modding the game by editing textures, arenas, and features, not by drawing, but through logic and the C# programming language. Apart from coding, I also learned Blender and game art. While practicing Blender, I ended up improving my 3D modeling and animation skills. Each experiment taught me how small changes in code could transform gameplay. This experience opened the door to a new set of skills for me.

People on YouTube claimed that this game could not be modded, so when I did it, I wanted everyone to know that it was indeed possible. I posted the mod on YouTube with zero expectations, and it gained around 10,000 views. This was the start of my YouTube journey. I built a community of over 4,000 subscribers while collaborating with other YouTubers like Young Fro, a YouTuber with 113,000 subscribers. My legacy in the community was further cemented when pro wrestler Zilla Fatu, who has around 120,000 subscribers, played my mod and praised it.

This experience changed my perception of creativity. I realized that coding itself is an art form that blends logic with imagination. Through modding, I learned how to think like both a developer and a gamer. It showed me how creative problem solving in technology can connect people across continents and turn curiosity into innovation.

Now, whenever I build or code something new, I remember that counselor's words. I have learned, creativity is not always painted; sometimes it is compiled too.

PIQ 3: What have you done to make your school or your community a better place?

When I built my waste-segregation app, I thought technology would change communities, but I was wrong. It's the people who change communities, and technology only helps them to start. I made Eco Bin with a simple thought: help families separate dry and wet waste using a multilingual, offline app. However, when I introduced it, people were hesitant. The elderly weren't comfortable using the app, and parents didn't want an extra step. Nothing changed. Instead of forcing the current system, I tried something different. I turned to the most open-minded group in society, children and teenagers. I asked them to take a selfie with their sorted waste and post it in the society's WhatsApp group. This started a chain reaction. The group was full of kids holding green and blue bins. Shortly, grandparents joined with enthusiasm, followed by parents. Within a span of one month, everyone started using the app. EcoBin took the shape I intended it to take: it became a community movement. It taught me that leadership is not about convincing, it's about inspiring. I carried this learning to my AI Literacy Drive. In a village near Nagpur, while researching AI awareness, I met a woman who displayed a lot of curiosity. She had studied only till fifth grade, yet her eyes sparkled when she watched me use an audio AI tool. She said in Marathi, "Mala pan AI shikaycha aahe," or I want to learn AI too. She kept calling AI "magical" whenever AI replied back to her in Marathi, the regional language of Maharashtra. Teaching her felt less like instruction and more like sowing a seed of empowerment. She went on to explain AI to her family. Watching this mother spread knowledge reminded me how powerful the role of community can be. Across both projects, I realized that community service is not all about solving problems but empowering people. Whether helping an urban society become more sustainable or introducing AI to a village, my purpose will be the same: use technology to empower, and spark confidence in those who never believed it was meant for them.

PIQ 4: Beyond what has already been shared in your application, what do you believe makes you a strong candidate for admissions to the University of California?

My forefathers came from Rajasthan, one of India's driest desert states. In a geography where temperatures varied between scorching and freezing, every drop of water and every grain of food was precious. From this dearth, they built abundance, turning resourcefulness into entrepreneurship and restraint into prosperity. For a Marwari from Rajasthan, "less is more" is not a belief. It is a survival instinct passed down through generations.

I grew up watching that same ideology at home. My grandfather honed his capabilities to such an extent that he would close deals even before others began their calculations. My grandmother managed things so efficiently at home that nothing ever went to waste. The lesson I learned from them was that real innovation lies not in having more but in doing more with less. When I started learning Artificial Intelligence, I saw that same guiding principle reflected in technology. A few words could complete hours of work in minutes. All that was needed was a mobile phone and the internet. I started to look at AI as a digital extension of my Marwari cultural lineage: an art of optimization and purpose.

AI saves time the way my ancestors saved resources. Work that once required hours of manual effort, like researching data, drafting documents, or performing repetitive operations, can now be finished in minutes. It is all about working smart, not hard. In this fast paced world where time is money and resources are limited, AI is bridging the gap between ideas and impact.

This perspective has changed my approach. I have used AI to help communities segregate waste, which helped transform disposed materials into usable resources, with Ecobin. I worked to make technology accessible for teachers and students without adding extra cost or complexity, with my Apple iPad Integration initiative at school.

I want to channel my unique heritage with technology to design systems that waste less, serve more, and work smartly. The world today is embracing what Marwaris have known for ages: sustainability, and incisive thinking. I bring this deeply traditional, successful approach to building a future where less truly becomes more.

UCAS Essays

Why do you want to study this course or subject?

I chose Computer Science because I believe it is the most transformative field since the internet. The impact of Artificial Intelligence extends beyond computing into fields like biology and astronomy. I have always been fascinated by Mathematics and Mechanical physics, grounded in vectors and calculus, showed me how complex systems can be made from simple, well defined laws. This interest led to practical exploration, including my Top 50 All India Rank in the Future Tech Olympiad by 1M1B and IBM, which tested my knowledge of technology and machine learning. During my internship at Lighthouse Learning, I further applied these ideas by developing a feature called "Future Me" for the Argus school app, using an open source Stable Diffusion model to help students imagine their aspirational careers. Working on this project taught me how theoretical concepts can be implemented into tools with real world impact. I want to contribute to this transformation and help drive progress for humanity by applying AI to solve real world problems and advance scientific discovery. Together, these experiences shaped Computer Science as a medium to explore problems, ask deeper questions, and find meaningful solutions.

How have your qualifications and studies helped you to prepare for this course or subject?

My passion for mathematics and computer science has given me the analytical skills required for Artificial Intelligence. These skills, combined with practical applications, have prepared me well to further study in AI. This knowledge has been ingrained in me by a strong foundation laid by the CBSE syllabus, where I extensively learned to use Python to develop logical reasoning and problem-solving skills. Principal topics such as calculus, probability, and programming helped me understand how mathematical concepts result in computational models used in AI. I further strengthened this foundation by completing a Number Theory and Cryptography course from the University of California, San Diego, which introduced me to mathematical structures and encryptions. I scored a 4 in AP Computer Science, which reinforced my understanding of programming and algorithmic maths. Additionally, my internship at Lighthouse Learning gave me practical experience working in teams, collaborating across positions, and applying AI and software concepts in real-world projects, which has prepared me well for the demands of an AI and computer science program.

What else have you done to prepare outside of education, and why are these experiences useful?

Outside formal education, I won the CREST Gold Award by developing EcoBin, an AI-powered offline waste segregation application. This app helps in sorting waste into wet, dry, or e-waste. I tried to identify a real-world problem, that is, waste segregation at source for better recycling and sustainability. This app not only saves time and effort but also resources. Through this process, I strengthened my ability to apply AI practically and responsibly.

EcoBin turned from a digital tool into a community-driven initiative, with over 100 families adopting waste segregation within a month through group participation rather than just using technology. Similarly, during my AI Literacy Drive in rural India, I observed how technology, when made accessible through local language and cultural settings, enabled first-time learners to engage confidently with AI and transfer knowledge within their families and community. These experiences reinforced my understanding that technology creates impact only when it empowers people.

Alongside this, as my school's inaugural AI Captain, I led the integration of iPads across fourth and fifth grades. I identified recurring technical issues and found user-friendly solutions. I established an iPad Buddy Team, providing peer-to-peer support that resolved over fifty issues and assisted more than thirty teachers in one month. I also created open-source resources to ensure scalability and a wider approach. All these experiences strengthened my interest in studying Computer Science for building solutions which are both scalable and sustainable.

New York University (9%)

Primary Area of Study: Computer Science (Accepted for)

Global Site Interest: I prefer to start in New York, but am also interested in London

Alternative School/Programs:

1. Computer and Data Science (College of Arts and Science)
2. Computer Engineering (Tandon School of Engineering)
3. Data Science (College of Arts and Science)
4. Mathematics (Tandon School of Engineering)

NYU Essay

During an AI Literacy Drive in Aamgaon, a village in India, I met people with different educational backgrounds and technological perspectives, to an extent that many were hearing about artificial intelligence for the first time. The biggest setback was communication with undereducated individuals. Training them to use smartphones in their regional language was a tough task, as the majority of users were unfamiliar with them. To bridge this gap, I used AI voice commands and translation tools to explain concepts and demonstrate applications. Explaining prompts and setting up tools on their phones required time and patience. Social barriers also emerged: many women did not participate, while children and teenagers were the most eager to learn. I focused on supporting learners and showing practical uses of AI, such as analyzing crops, understanding climate patterns, or solving everyday problems. As a few participants gained confidence, learning became impactful. Following this, women began to participate. I realized that women not only learned quickly but also were teachers in their families. For me, learning and technology were always accessible, but in this village, I saw how education itself is a privilege, and access to technology can feel like a dream. I learned that people do not lack ability; they often lack the right tools, guidance, and support. By bridging educational and social gaps, I learned the importance of patience, inclusion, and empathy. This experience taught me what it truly means to be a bridge builder that is, connecting people to grow and learn together.